01 / A BUSINESS QUESTION WITH A MEASUREMENT BOUNDARY

Price the trip. Then price the missing time.

Should a yellow-taxi operator favor airport-to-Manhattan trips over trips within Manhattan?

10.15m
selected-route trips
January-April 2025

$83.4/h
April leader
LaGuardia to Manhattan

9.3 min
extra time to parity*
April scenario

RECOMMENDED DECISION

Fund a small time-measurement pilot before changing dispatch priorities.

The April occupied-time comparison favors **LaGuardia to Manhattan**: $83.4 per occupied hour versus $64.3 for Manhattan trips (1.30x). That comparison omits the time needed to obtain the trip and return to service.

METRIC CONTRACT

Fare per occupied hour = total fare_amount / total occupied minutes x 60

A duration-weighted ratio of sums. Fare excludes tips, tolls and separate surcharges. It is a trip productivity measure; full-shift income and profit require additional time and cost data.

POPULATION & ELIGIBILITY

10,148,555 study trips / 15,168,579 source records. Pickups: 01 Jan-30 Apr 2025. Jan-Mar calibrates; April checks. Eligibility: card/cash; rate codes 1/2; duration 1-180 minutes; distance above 0 and at most 100 miles; base fare above $0 and at most $500; known zones; pickup month matches file. Trips touching 09 Mar 2025 are excluded for clock-change ambiguity.

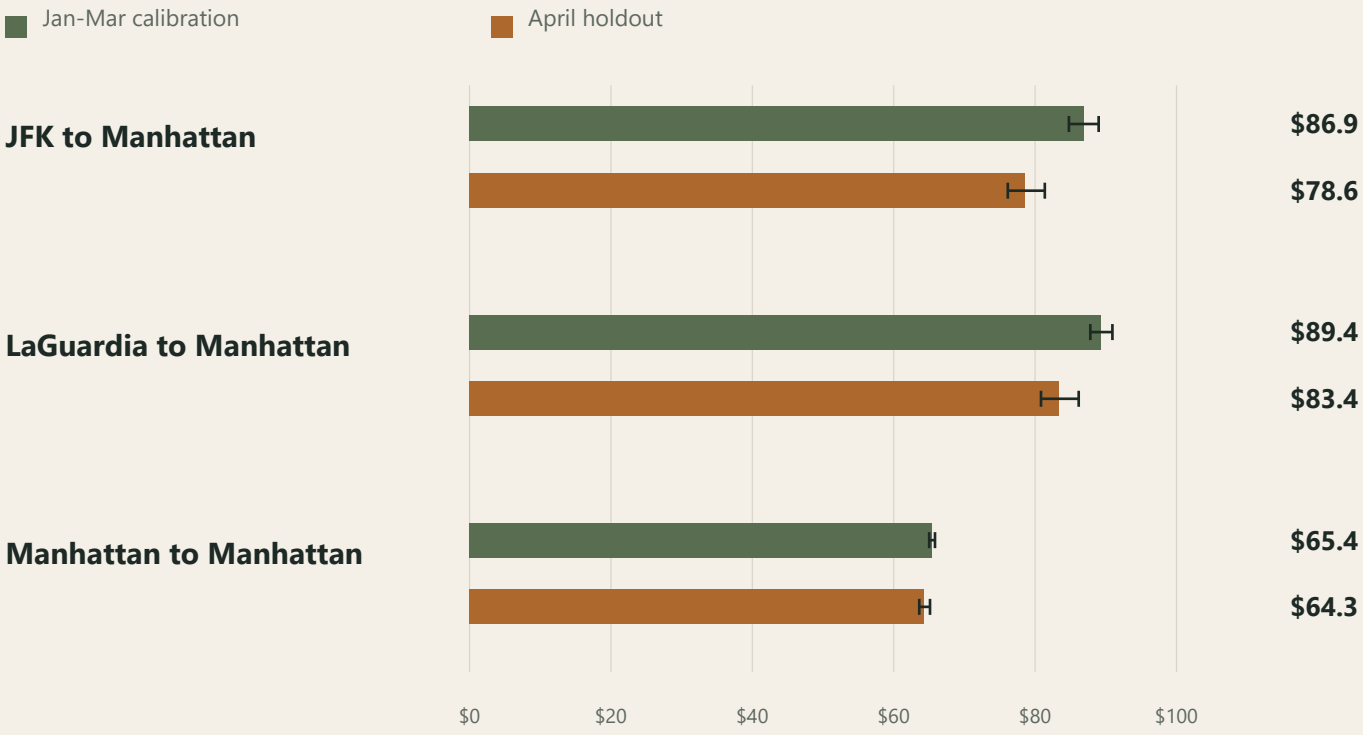
*Parity assumes zero unobserved time for the Manhattan benchmark; airport extra minutes are assumed, not recorded. Source: NYC TLC, Yellow Taxi Trip Records, Jan-Apr 2025. Source links and limits on page 3.

02 / HOLD THE QUESTION FIXED, CHECK A LATER MONTH

Does the route ranking survive April?

The comparison uses base fare divided by occupied time. Error bars resample whole calendar days.

FARE PER OCCUPIED HOUR / USD



DENOMINATORS / TRIPS AND OCCUPIED HOURS

ROUTE	JAN-MAR	APRIL
JFK to Manhattan	197,073 trips 159,139 occupied hours	75,303 trips 67,222 occupied hours
LaGuardia to Manhattan	186,178 trips 88,351 occupied hours	71,029 trips 36,980 occupied hours
Manhattan to Manhattan	7,087,136 trips 1,394,958 occupied hours	2,531,836 trips 528,538 occupied hours

The complete route ranking is preserved in April. This is a descriptive temporal check. Trip mix, hour of day and congestion can still explain differences between routes.

95% percentile intervals: 2,000 calendar-day bootstrap replicates, separately by period. Intervals describe day-to-day variation under the resampling assumption; they do not establish a dispatch-policy effect.

03 / TURN AN UNOBSERVED INPUT INTO A TESTABLE DECISION

How much missing time erases the premium?

A sensitivity calculation exposes the operating condition that the trip records cannot measure.

JFK TO MANHATTAN

11.9 min

city added time = 0 min per trip
If city adds 10 min: **64.2 min**

LAGUARDIA TO MANHATTAN

9.3 min

city added time = 0 min per trip
If city adds 10 min: **41.6 min**

Scenario: hold April fare and occupied time fixed; add the same unobserved minutes to every airport trip. The Manhattan comparator remains $64.3/h with **zero** added time. These break-even values are sensitivity thresholds, not estimates of real queues or a prediction of earnings.

PROPOSED PILOT / OPERATIONS + ANALYTICS

01 **Measure the whole cycle**

Capture queue, repositioning, idle and occupied minutes alongside fare and actual operating costs. Define the start and end of each duty block.

02 **Compare like operating windows**

Randomize feasible dispatch strategies within comparable time blocks, with driver participation. Record weather, day and congestion; set a sample-size plan after baseline measurement.

03 **Use an explicit decision gate**

Primary: base fare per full duty-hour. Separately assess cost-adjusted contribution and service quality. Adopt a policy only after the measured benefit clears a pre-agreed threshold.

Evidence boundary. Historical filtered trips are not a random policy experiment. Public records lack a linked full-shift activity log, queue time and operating expenses. Day-level intervals cannot remove selection bias. TLC does not warrant record accuracy.

Source: nyc.gov/site/tlc/about/tlc-trip-record-data.page
Yellow Taxi Data Dictionary (18 Mar 2025) | NYC terms of use | Jan-Apr 2025 Parquet files; source hashes recorded in the project manifest.